

**ADVANCED MICROECONOMETRICS**  
**Problem Set - Day 3**

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1. Assume the following economic framework:

$$Y_1 = \alpha_0 + \varphi + \beta_1 X + U_1 \quad (1)$$

$$Y_0 = \alpha_0 + \beta_0 X + U_0 \quad (2)$$

where  $U_0$  and  $U_1$  are unobserved variables,  $X$  represents observed controls, and  $\varphi$  represents the average benefit of the treatment ( $D = 1$ ). Individuals decide whether or not to receive treatment ( $D = 1$  or  $D = 0$ ) based on a latent variable  $I$  :

$$I = \gamma'Z - V \quad (3)$$

where  $Z$  and  $V$  represent observed and unobserved variables, respectively. Thus, we can define a binary variable  $D$  indicating treatment status:

$$D = \begin{cases} 1 & \text{if } I > 0 \\ 0 & \text{if } I \leq 0 \end{cases} \quad (4)$$

Finally, assume error terms in the model are not independent even conditioning on observables, i.e.  $U_1 \not\perp U_0 \not\perp V \mid (Z, X)$ , but  $(U_1, U_0, V) \perp\!\!\!\perp (Z, X)$ .

- (a) Present expressions for the following treatment parameters: Average Treatment Effect, Treatment on the Treated, Treatment on the Untreated, Marginal Treatment Effect, Instrumental Variables, OLS, and LATE. Show how these treatment parameters can be expressed as a weighted average of the Marginal Treatment Effect.

Suppose now that the error terms in the model have the following structure:

$$\begin{aligned} U_1 &= \sigma_1 \epsilon \\ U_0 &= \sigma_0 \epsilon \\ V &= \sigma_V^* \epsilon \\ \epsilon &\sim N(0, 1) \end{aligned}$$

- (b) Present the closed form expressions for the Treatment on the Treated and Treatment on the Untreated in this case.
- (c) Define the propensity score in this model.
- (d) Incorporate the propensity score ( $\Pr(D = 1 \mid Z = z)$ ) in your definition of the Marginal Treatment Effect, Treatment on the Treated, Treatment on the Untreated and LATE.

Now add more structure to the problem. In particular, assume:

$$\begin{aligned}
 \sigma_1 &= 0.012 & \sigma_0 &= -0.05 & \sigma_V^* &= -1 \\
 \alpha_0 &= 0.02 \\
 \beta_0 &= 0.5 & \beta_1 &= 0.8 \\
 \varphi &= 0.2 \\
 \gamma_0 &= 0.2 & \gamma_1 &= 0.3 \\
 X &\sim N(2,4) & Z &\sim N(-1,9)
 \end{aligned}$$

- (e) Generate a random samples of 5000 observations. Present the OLS estimator of the “effect” of the treatment effect from:

$$Y_i = \pi_0 + \pi_1 X + \pi_2 D + \epsilon_i$$

Can you identify ATE from this regression model?

- (f) Decompose the OLS estimator as the sum of  $E[Y_1 - Y_0 \mid X, D = 1]$  and sorting bias.
- (g) Implement the propensity score matching estimator. Compare your result with those from OLS.
- (h) Plot the Marginal Treatment Effect and the Average Treatment effect against  $u_D$  ( $F_V(V) = U_D$  where  $F_V$  represents the CDF of a standard normal random variable). Provide a brief explanation of your results.

Now modify the decision model assuming the presence of two instruments :  $\gamma_0 = 0.2$ ,  $\gamma_1 = 0.3$ ,  $\gamma_2 = 0.1$  with  $Z_1 \sim N(-1, 9)$ ,  $Z_2 \sim N(1, 9)$  where  $Z_1 \perp\!\!\!\perp Z_2$ .

- (i) Discuss which parameters of the model can be identified ( $\alpha_0, \varphi, \gamma$ , etc.) under normality.
- (j) Generate 100 random samples of 5000 observations. For each sample compute the parameters discussed in part (a). In the particular case of LATE you can select any pairs of values of the propensity score. Finally, compute the average and standard deviations (across samples) of each of these parameters.
- (k) Present a figure of the estimated Marginal Treatment Effect (including its confidence interval) against its actual value. Notice that both objects are plotted against  $u_D$ .
- (l) Finally, compare your estimates of TT, TUT and LATE with what you obtain using instrumental variables. Present the IV estimates when  $Z$  and the propensity score are used as instruments. Why would you prefer the propensity score?
- (m) When would you use the IV estimator to compute the “treatment effect ” of  $D$ ?